outdoors_root_growth_statistics

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Load packages

```
library(tidyverse)
library(lme4)
library(lmerTest)
library(emmeans)
library(car)
library(ggplot2)
library(ggpubr)

dir.create("Outdoors root growth results", showWarnings = FALSE)
```

Import data

```
df <- read.csv("outdoors_root_growth_data.csv")
```

Prepare variables

```
df <- df %>%
  mutate(
    Experiment = factor(Experiment),
    Species = factor(Species),
    Treatment = factor(Treatment)
  )
```

Anova model

```
options(contrasts = c("contr.sum", "contr.poly"))

model <- lm(New.growth.sum ~ Treatment * Species + Experiment,
   data = df)

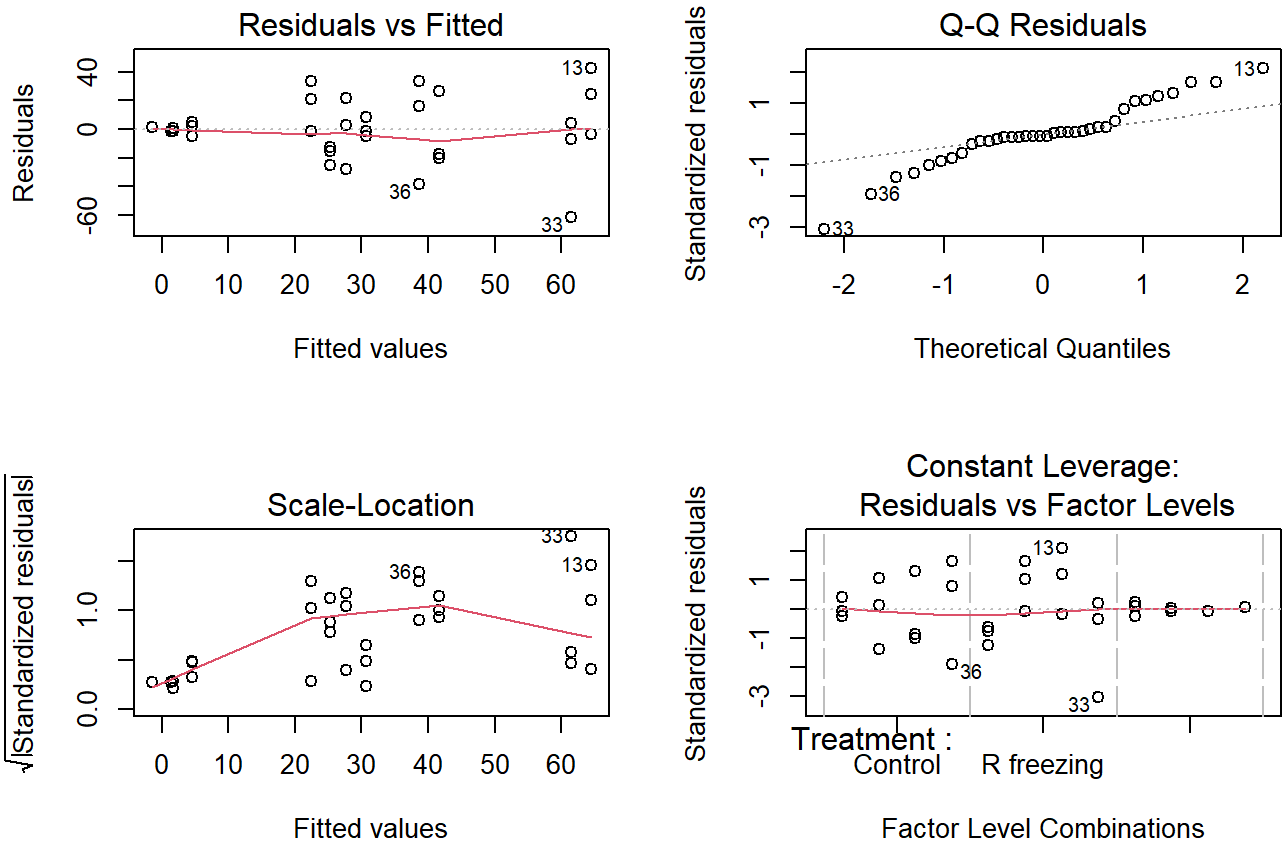
anova_results <- Anova(model, type = 3)
anova_results
```

```
## Anova Table (Type III tests)
##
## Response: New.growth.sum
##                    Sum Sq Df F value    Pr(>F)
## (Intercept)       25392.5  1 50.2838 8.408e-08 ***
## Treatment         11703.3  2 11.5878 0.0002002 ***
## Species            2204.5  1  4.3655 0.0455568 *
## Experiment           78.4  1  0.1553 0.6964345
## Treatment:Species  2784.3  2  2.7568 0.0801520 .
## Residuals         14644.5 29
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
write.csv(
    anova_results,
    "Outdoors root growth results/ANOVA_table.csv")
```

Diagnostics

```
par(mfrow = c(2,2))
plot(model)
```



Estimated marginal means

```
emm <- emmeans(model, pairwise ~ Treatment | Species)

pairs_df <- as.data.frame(emm$contrasts)
pairs_df
```

```
## Species = D. flexuosa:
##  contrast                    estimate       SE df t.ratio p.value
##  Control - R freezing         5.39188 12.97413 29   0.416  0.9095
##  Control - (R+S freezing)    26.10490 12.97413 29   2.012  0.1273
##  R freezing - (R+S freezing) 20.71302 12.97413 29   1.596  0.2632
##
## Species = E. vaginatum:
##  contrast                    estimate       SE df t.ratio p.value
##  Control - R freezing       -22.86350 12.97413 29  -1.762  0.2001
##  Control - (R+S freezing)    40.14391 12.97413 29   3.094  0.0117
##  R freezing - (R+S freezing) 63.00740 12.97413 29   4.856  0.0001
##
## Results are averaged over the levels of: Experiment
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```
write.csv(pairs_df,
          "Outdoors root growth results/Treatment_Pairwise_Comparisons.csv",
          row.names = FALSE)
```